

Lecture 4: Linear Time Selection/Median

Jessica Sorrell

September 10, 2026

601.433/633 Introduction to Algorithms

Slides by Michael Dinitz

Announcements

Office hours:

- ▶ I will have office hours every Tuesday 4:30 - 5pm (start next week).
- ▶ Prof. Dinitz will have office hours every Thursday 11 - 12 (started today).
- ▶ Shruthi will hold recitations every Friday 12–1 in Malone 222 (starting tomorrow).
 - ▶ Small room — will change if attendance high.
- ▶ Normal CA office hours M-Th 6pm–9pm (starting next week). Still figuring out rooms.

HW1 out now! Due next Thursday by 11:59pm.

Intro and Problem Definition

Last few lectures: sorting

- ▶ Insertion sort, quicksort: $O(n^2)$
- ▶ Mergesort, heapsort: $O(n \log n)$

Today: two related problems

- ▶ Median: Given array $\mathbf{A}$ of length n , find the median: $\lceil n/2 \rceil$ 'nd smallest element.
- ▶ Selection: Given array $\mathbf{A}$ of length n and $k \in [n] = \{1, 2, \dots, n\}$, find k 'th smallest element.

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Can solve both in $O(n \log n)$ time via sorting. Faster?

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- ▶ Need to keep track of $n/2$ smallest.
- ▶ When scanning, for each element seen, determine if it is among the k smallest. If so, remove the previous max of the $n/2$ elements tracked.
 - ▶ Not easy to do! Foreshadow: would need to use a *heap*. $\Theta(\log n)$ worst-case time.

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- ▶ $\Theta(n \log n)$ worst-case time.

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2. Pick a pivot element $\mathbf{p}$ (somehow).
3. Compare each element of $\mathbf{A}$ to $\mathbf{p}$, creating subarrays $\mathbf{L}$ of elements less than $\mathbf{p}$ and $\mathbf{G}$ of elements greater than $\mathbf{p}$.
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Sketch here: good exercise to do at home!

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Prove by induction (“loop invariant”) that on any call to $\text{Quickselect}(\mathbf{X}, \mathbf{a})$, the element we’re looking for is $\mathbf{a}$ ’th smallest of $\mathbf{X}$.

- ▶ Base case: first call to $\text{Quickselect}(\mathbf{A}, \mathbf{k})$. Correct by definition.
- ▶ Inductive case: suppose the element we’re looking for is the $\mathbf{b}$ ’th smallest element of $\mathbf{Y}$ when we call $\text{Quickselect}(\mathbf{Y}, \mathbf{b})$.
 - ▶ If we return element: correct
 - ▶ If we recurse on $\mathbf{L}$: correct
 - ▶ If we recurse on $\mathbf{G}$: correct

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- ▶ $p = A[n/2]$: fun example, work it out at home!

Linear-time Version

Intuition:

- ▶ Simple pivot rule (first element, last element, etc.) bad because might not split array well.
- ▶ Can't find pivot that's exactly in the middle: median problem!
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Median-of-medians:

- ▶ Split **A** into **$n/5$** groups of **5** elements each.
- ▶ Compute median of each group.
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Claim: **p** is a good pivot, and can find **p** efficiently.

Median-of-Medians is good pivot

Theorem

$|L|$ and $|G|$ are both at most **$7n/10$** when **p** is the median of medians.

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Finding Median of Medians

Have $n/5$ elements (median of each group). Want to find their median.

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Recursion! Use the same algorithm on array of medians.

BFPRT

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BFPRT(A, k)

1. If $|A| \leq 5$, sort A and return the k th element.
2. Group A into $n/5$ groups of 5, and let A' be an array of size $n/5$ containing the median of each group.
3. Let $p = \text{BFPRT}(A', n/10)$, i.e., recursively find the median p of A' (the median-of-the-medians).

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4. Split A using p as a pivot into L and G .
5. Recurse on the appropriate piece:
 - 5.1 if $|L| = k - 1$ then return p .
 - 5.2 if $|L| > k - 1$ then return $\text{BFPRT}(L, k)$.
 - 5.3 if $|L| < k - 1$ then return $\text{BFPRT}(G, k - |L| - 1)$.

BFPRT Analysis

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- ▶ Step 4 (recurse):

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Let $T(n)$ be the (worst-case) running time on an array A of size n .

- ▶ Step 1 (grouping, find medians): $O(n)$ time
- ▶ Step 2 (find median of medians): $T(n/5)$ time
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$$T(n) \leq T(7n/10) + T(n/5) + cn$$

Guess $T(n) \leq 10cn$:

$$T(n) \leq 10c(7n/10) + 10c(n/5) + cn = 9cn + cn = 10cn$$

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Let $T(n)$ be time on input of size n .

- ▶ BFPRT to find pivot takes $O(n)$ time
- ▶ Splitting around pivot takes $O(n)$ time
- ▶ Each recursive call takes $T(n/2)$ time

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$$T(n) = 2T(n/2) + cn \implies T(n) = \Theta(n \log n)$$